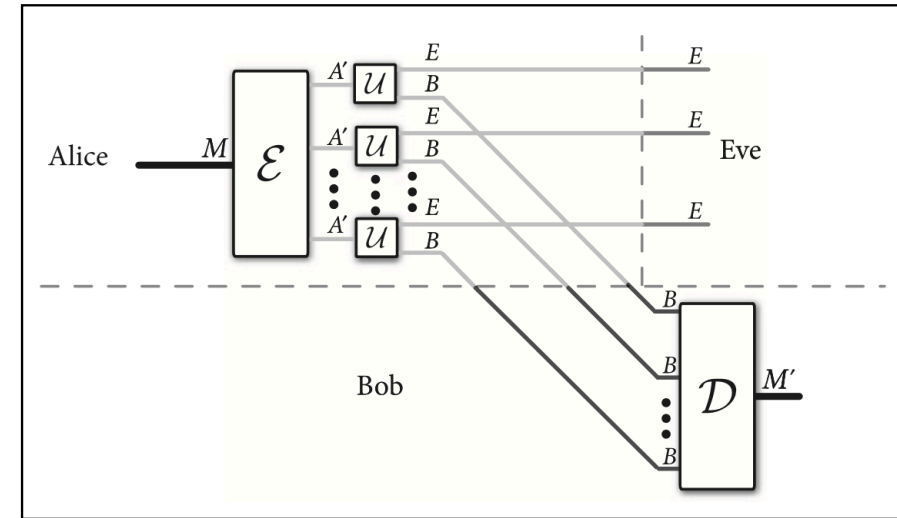




Private Classical Capacity

- Alice encodes some private message m into a quantum codeword $\rho_{A'n}^m$ and transmits it over many uses of a quantum channel.
- The goal is for Bob to be able to reliably distinguish the message, while the channel's environment Eve should not be able to learn anything about it.
- Eve Channel: $\hat{\mathcal{N}}_{A' \rightarrow E}(\sigma) = \text{Tr}_B (U \sigma U^\dagger)$, where $U = U_{A' \rightarrow BE}^{\mathcal{N}}$ is an isometric extension of $\mathcal{N}_{A' \rightarrow B}$



- **Private Capacity:** $P(\mathcal{N}) \geq \max[I(X; B)_\sigma - I(X; E)_\sigma]$ where the maximization is taken over classical-quantum states $\rho_{XA'} = \sum_x p_X(x) |x\rangle\langle x|_X \otimes \rho_{A'}^x$ and $\sigma_{XBE} = U_{A' \rightarrow BE}^{\mathcal{N}} (\rho_{XA'}) := \sum_x p_x |x\rangle\langle x|_X \otimes U \rho_{A'}^x U^\dagger$
- Used in Quantum Key Distribution Secret Key Rate Analysis

Private Capacity of Identity Channel



11-Jul-2026

$|0\rangle \rightarrow |0\rangle, |e\rangle$
 $|1\rangle \rightarrow |1\rangle, |e\rangle$

$A \xrightarrow{P} B$
 $A \xrightarrow{P} E$

10×01
 10×01

Ex: Private Capacity of Identity Channel
 $\mathcal{N} = \text{id}_2$

Isometry Channel: $U: A \rightarrow BE$
 $U|\psi\rangle_A = |\psi\rangle_B \otimes |0\rangle_E$

Complementary: $\mathcal{N}'(\rho) = |0\rangle\langle 0|_E$

Channel

$\Rightarrow$ Eve does not receive any information.

Private Capacity = $\max_{\rho} \{ I(X; B) - I(X; E) \}$

ρ

$A \xrightarrow{P} B$

E

B

Eve's state is fixed, & $I(X; E) = 0$

$P(\mathcal{N}) = \max_{\rho} I(X; B) \rightarrow \text{Holevo Capacity of identity channel}$
 $= 1 \text{ bit/ch. use.}$

$$I(X; B) \leq S(\rho_B) \leq \log 2 = 1.$$

$$\begin{matrix} |0\rangle \\ \frac{1}{\sqrt{2}} \end{matrix}, \begin{matrix} |1\rangle \\ \frac{1}{\sqrt{2}} \end{matrix} \rightarrow \rho_B = I/2 \rightarrow S(\rho_B) = 1.$$

$$I(X; B) = 1$$

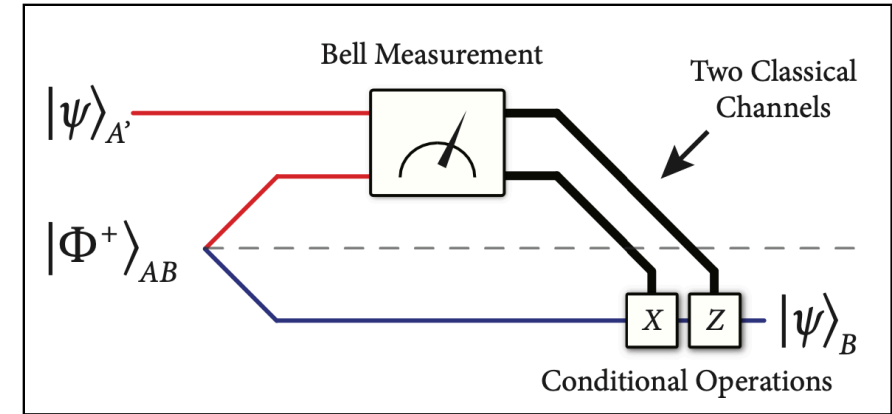


Quantum Capacity



Quantum Teleportation

- Alice would like to transmit an arbitrary quantum state $|\psi\rangle\langle\psi|_{A'}$ to Bob.
- Alice and Bob share an ebit before the protocol begins.
- Alice can “teleport” her quantum state to Bob by consuming the entanglement and two uses of a noiseless classical bit channel.
- Algebraic Trick: Let $|\psi\rangle\langle\psi|$ be the state of a qubit where $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.



- Using the algebra of the tensor product, can show that:
 $|\psi\rangle_{A'}|\Phi^+\rangle_{AB} \propto |\Phi^+\rangle_{A'A}|\psi\rangle_B + |\Psi^+\rangle_{A'A}X_B|\psi\rangle_B + |\Phi^-\rangle_{A'A}Z_B|\psi\rangle_B + |\Psi^-\rangle_{A'A}X_BZ_B|\psi\rangle_B$
 $|\psi\rangle_{A'}|\Phi^+\rangle_{AB} \propto |\Phi^+\rangle_{A'A}|\psi\rangle_B + X_A|\Phi^+\rangle_{A'A}X_B|\psi\rangle_B + Z_A|\Phi^+\rangle_{A'A}Z_B|\psi\rangle_B + Z_AX_A|\Phi^+\rangle_{A'A}X_BZ_B|\psi\rangle_B$
- Performing the Bell measurement channel on systems AA' leads to the following state:
 $\frac{1}{4} [|00\rangle\langle 00|_{AA'} \otimes |\psi\rangle\langle\psi|_B + |01\rangle\langle 01|_{AA'} \otimes X_B|\psi\rangle\langle\psi|_B X_B + |10\rangle\langle 10|_{AA'} \otimes Z_B|\psi\rangle\langle\psi|_B Z_B + |11\rangle\langle 11|_{AA'} \otimes X_BZ_B|\psi\rangle\langle\psi|_B Z_BX_B]$
- Alice then sends the two classical bits in AA'' to Bob. Bob can then undo the Pauli rotations and recover the state $|\psi\rangle\langle\psi|_B$

$$\begin{aligned} |00\rangle_{A'A} &= \frac{1}{\sqrt{2}} (|\Phi^+\rangle + |\Phi^-\rangle) , |11\rangle_{A'A} = \frac{1}{\sqrt{2}} (|\Phi^+\rangle - |\Phi^-\rangle) \\ |01\rangle_{A'A} &= \frac{1}{\sqrt{2}} (|\Psi^+\rangle + |\Psi^-\rangle) , |10\rangle_{A'A} = \frac{1}{\sqrt{2}} (|\Psi^+\rangle - |\Psi^-\rangle) \end{aligned}$$

$$\text{BSM: } |\Phi^+\rangle \rightarrow 00 , |\Phi^-\rangle \rightarrow 10 , |\Psi^+\rangle \rightarrow 01 , |\Psi^-\rangle \rightarrow 11$$

Teleportation Derivation



Teleportation Protocol

$$|\psi\rangle_{A'} = \alpha|0\rangle + \beta|1\rangle$$

$$|\psi\rangle_{A'AB} = \underbrace{(\alpha|0\rangle + \beta|1\rangle)}_{|\psi\rangle_{A'}} \otimes \underbrace{\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)}_{|\phi^+\rangle_{AB}}$$

$$= \frac{1}{\sqrt{2}} (\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle)_{A'AB}$$

$$|00\rangle_{A'A} = \frac{1}{\sqrt{2}} (|\phi^+\rangle + |\phi^-\rangle)_{A'A} \quad \left\{ \begin{array}{l} |\phi^+\rangle, |\phi^-\rangle \\ |\psi^+\rangle, |\psi^-\rangle \end{array} \right\}$$

$$|01\rangle_{A'A} = \frac{1}{\sqrt{2}} (|\psi^+\rangle + |\psi^-\rangle)_{A'A}$$

$$|11\rangle_{A'A} = \frac{1}{\sqrt{2}} (|\phi^+\rangle - |\phi^-\rangle)_{A'A}$$

$$|10\rangle_{A'A} = \frac{1}{\sqrt{2}} (|\psi^+\rangle - |\psi^-\rangle)_{A'A}$$

$$\rightarrow \frac{1}{\sqrt{2}} \left(\underbrace{\frac{\alpha}{\sqrt{2}} (|\phi^+\rangle + |\phi^-\rangle)}_a |0\rangle + \underbrace{\frac{\alpha}{\sqrt{2}} (|\psi^+\rangle + |\psi^-\rangle)}_b |1\rangle + \underbrace{\frac{\beta}{\sqrt{2}} (|\psi^+\rangle - |\psi^-\rangle)}_a |0\rangle + \underbrace{\frac{\beta}{\sqrt{2}} (|\phi^+\rangle - |\phi^-\rangle)}_b |1\rangle \right)$$

$$\frac{1}{2} \left(|\phi^+\rangle_{A'A} \underbrace{(\alpha|0\rangle + \beta|1\rangle)}_{|\psi\rangle_B} + |\phi^-\rangle_{A'A} \underbrace{(\alpha|0\rangle - \beta|1\rangle)}_{|\psi\rangle_B} + |\psi^+\rangle_{A'A} \underbrace{(\alpha|1\rangle + \beta|0\rangle)}_{|\psi\rangle_B} + |\psi^-\rangle_{A'A} \underbrace{(\alpha|1\rangle - \beta|0\rangle)}_{|\psi\rangle_B} \right)$$

$$\frac{1}{2} \left(|\phi^+\rangle_{A'A} |\psi\rangle_B + |\phi^-\rangle_{A'A} Z_B |\psi\rangle_B + |\psi^+\rangle_{A'A} X_B |\psi\rangle_B + |\psi^-\rangle_{A'A} Z_B X_B |\psi\rangle_B \right)$$

BSM on A'A

$\frac{1}{4}$	$ \phi^+\rangle$	$\rightarrow$	00	$\rightarrow$	$ \psi\rangle_B$
$\frac{1}{4}$	$ \phi^-\rangle$	$\rightarrow$	10	$\rightarrow$	$Z_B \psi\rangle_B$
$\frac{1}{4}$	$ \psi^+\rangle$	$\rightarrow$	01	$\rightarrow$	$X_B \psi\rangle_B$
$\frac{1}{4}$	$ \psi^-\rangle$	$\rightarrow$	11	$\rightarrow$	$Z_B X_B \psi\rangle_B$

$$Z_B Z_B = I, \quad X_B X_B = I \quad \& \quad (Z_B X_B)(Z_B X_B) = I$$

Quantum Teleportation Interpretation



- Alice can't obtain the parameters α, β of the qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.
- **No Violation of “No-Cloning Theorem”**: State vanishes at Alice's side and appears at Bob's location
- Does not violate the “**No faster than light**” communication principle: A classical communication is necessary between Alice and Bob to correctly recover the input qubit